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Experiment Title : Digital Modulation Techniques: Generation and Observation of ASK, FSK, and PSK Signals

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Experiment No. 10

Title: Digital Modulation Techniques: Generation and Observation of ASK, FSK, and PSK Signals

Aim

To design and simulate Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK), and Binary Phase Shift Keying (BPSK) circuits in LTspice using a PULSE voltage source as the binary message signal and Behavioral Voltage Sources (BV) as modulators. To observe the modulated waveforms in the time domain and verify the modulation behaviour for each technique.

Software Used

LTspice 26.0.1 (High-performance SPICE simulation software).

Theory

1. What is Digital Modulation?

At its core, digital modulation modifies one or more properties of a high-frequency sinusoidal carrier wave—**amplitude, frequency, or phase**—based on the digital baseband signal.

Why is it used?

- **Efficiency:** It allows for better bandwidth utilization.
- **Security:** Digital signals are easier to encrypt.
- **Noise Immunity:** Digital systems are generally more robust against noise and interference than analog systems because they only need to distinguish between discrete states.

- **Multiplexing:** It enables multiple signals to be sent over the same medium simultaneously.

2. Amplitude Shift Keying (ASK)

Amplitude Shift Keying (ASK) is a form of digital modulation where the amplitude of the carrier wave is varied to represent binary data, while the frequency and phase remain constant.

- **How it works:** In its simplest form, **On-Off Keying (OOK)**, the carrier is transmitted to represent a binary '1' and switched off to represent a binary '0'.
- **Mathematical Representation:**

$s(t) = A_i \cos(2 \pi f_c t)$

Where A_i takes different discrete values depending on the bit.

- **Pros/Cons:** It is very simple to implement but highly susceptible to atmospheric noise and signal fading, as noise primarily affects the amplitude of a signal.

3. Frequency Shift Keying (FSK)

Frequency Shift Keying (FSK) involves shifting the frequency of the carrier wave between two or more discrete values.

- **How it works:** In **Binary FSK (BFSK)**, a binary '1' is represented by a high frequency (f_1), and a binary '0' is represented by a lower frequency (f_0).
- **Mathematical Representation:**

$s(t) = A \cos(2 \pi f_i t)$

Where f_i switches between the mark frequency (1) and space frequency (0).

- **Pros/Cons:** FSK is much more resistant to noise than ASK because noise rarely changes the frequency of a wave. However, it requires more bandwidth than ASK or BPSK.

4. Binary Phase Shift Keying (BPSK)

Binary Phase Shift Keying (BPSK) is a modulation scheme where the phase of the carrier wave is shifted to represent different digital states.

- **How it works:** In BPSK, the phase of the carrier is shifted by 180 degrees (or π radians) to distinguish between a '0' and a '1'. For example, '1' might start at 0 degrees, while '0' starts at 180 degrees.
- **Mathematical Representation:**

$s(t) = A \cos(2\pi f_c t + \Phi_n)$

Where Φ_n is 0 or π.

- **Pros/Cons:** BPSK is very power-efficient and highly robust against noise. It is widely used in deep-space telemetry and satellite communications.

5. Significance of Modulation Parameters

To evaluate the performance and efficiency of these schemes, we look at three critical parameters:

Modulation Index (h)

The modulation index (specifically in FSK) describes how much the frequency of the carrier varies relative to the bit rate.

- **Significance:** It determines the bandwidth of the signal and the "distance" between the frequencies. A higher index provides better error performance (noise resistance) but consumes more bandwidth.

Deviation Ratio (D)

Similar to the modulation index, the deviation ratio is the ratio of the maximum frequency deviation to the maximum modulating frequency.

- **Significance:** In digital systems, it helps in defining the spectral width of the FSK signal. It is a key factor in designing filters for the receiver to ensure signal clarity.

Modulation Order (M)

Modulation order refers to the number of different symbols or states used to represent data. For example, in BPSK, M=2 (two phases). In 16-QAM, M=16.

- **Significance:**
    - **Data Rate:** Higher M allows you to send more bits per symbol (n = log2 M). For instance, M=4(QPSK) sends 2 bits per symbol.
    - **Bandwidth Efficiency:** Higher order modulation increases the data rate within the same bandwidth.
    - **Trade-off:** As M increases, the symbols get "closer" together, making the system more sensitive to noise and requiring a higher Signal-to-Noise Ratio (SNR).

Mathematical Representation

1. Amplitude Shift Keying (ASK)

In ASK, the amplitude of the carrier is switched between two levels depending on the binary message bit. For OOK (On-Off Keying), the simplest form of ASK:

$$s_{ASK}(t) = m(t) \times Ac \sin(2\pi fct) = V_{msg} * V_{carrier}$$

where $m(t) \in \{0, 1\}$ is the binary message (PULSE source), Ac = 1 is the normalised carrier amplitude, and fc is the carrier frequency.

- When m(t) = 1 (bit-1): carrier is transmitted.
- When m(t) = 0 (bit-0): output is zero.

2. Frequency Shift Keying (FSK)

In FSK, the frequency of the carrier is switched between two frequencies depending on the binary message bit:

$s_{FSK}(t) = Ac \sin(2\pi f_1 t)$ when m(t) = 0 (space frequency)

$s_{FSK}(t) = Ac \sin(2\pi f_2 t)$ when m(t) = 1 (mark frequency)

where f_1 = 50 kHz (space, bit-0) and f_2 = 100 kHz (mark, bit-1). The BV uses an IF() condition to select between the two SINE sources.

3. Binary Phase Shift Keying (BPSK)

In BPSK, the phase of the carrier is shifted by 180° depending on the binary message bit:

$$sBPSK(t) = d(t) \times Ac \sin(2\pi fct)$$

where $d(t) \in \{-1, +1\}$ is the bipolar version of the binary message. Since the PULSE source produces $m(t) \in \{0, 1\}$, the BV maps it to d(t) using:

$$d(t) = 2 \times m(t) - 1$$

Thus: d(t) = +1 when m(t) = 1 (bit-1, 0° phase), and d(t) = −1 when m(t) = 0 (bit-0, 180° phase reversal).

Circuit Description and Component Mapping

Common Components (used in all three circuits)

A single PULSE voltage source (Vmsg) serves as the binary message signal for all three modulations. It produces a 0V/1V square wave at a bit rate of 1 kbps (bit period Tb = 1 ms).

ASK — Component Table

Component	Source Type	Signal Produced	Key Setting
Vc	SINE source	Carrier: $\sin(2\pi \cdot fc \cdot t)$	Amplitude = 1V, Freq = 20k, DC offset = 0
Vmsg	PULSE source	Binary message: $m(t) \in \{0V, 1V\}$	Vinitial=0, Von=1, Trise=10n, Tfall=10n, Ton=0.5m, Tperiod=1m
B1	Behavioral Voltage (BV)	ASK output: V(msg) × V(carrier)	V = V(msg) * V(carrier)
R1	Resistor	50Ω output load	Value = 50

FSK — Component Table

Component	Source Type	Signal Produced	Key Setting
Vf1	SINE source	Space carrier: $\sin(2\pi \cdot f_1 \cdot t)$	Amplitude = 1V, Freq = 10k, DC offset = 0
Vf2	SINE source	Mark carrier: $\sin(2\pi \cdot f_2 \cdot t)$	Amplitude = 1V, Freq = 20k, DC offset = 0
Vmsg	PULSE source	Binary message: $m(t) \in \{0V, 1V\}$	Same as ASK: Ton=0.5m, Tperiod=1m

Component	Source Type	Signal Produced	Key Setting
B1	Behavioral Voltage (BV)	FSK output: selects f1 or f2	V = IF(V(msg)>0.5, V(f2), V(f1))
R1	Resistor	50Ω output load	Value = 50

PSK — Component Table

Component	Source Type	Signal Produced	Key Setting
Vc	SINE source	Carrier: sin(2π·fc·t)	Amplitude = 1V, Freq = 20k, DC offset = 0
Vmsg	PULSE source	Binary message: m(t) ∈ {0V, 1V}	Same as ASK: Ton=0.5m, Tperiod=1m
B1	Behavioral Voltage (BV)	BPSK output: phase-flipped carrier	V = V(carrier) * (2*V(msg) − 1)
R1	Resistor	50Ω output load	Value = 50

Simulation Procedure

Three separate circuits are built — one each for ASK, FSK, and PSK. The PULSE source Vmsg is identical in all three circuits.

Part A: ASK Circuit

Step A1: ***Place the Carrier Source Vc (SINE, 20 kHz, Vamp = 1)***

- Select voltage source and place it on the schematic.
- Right-click on the voltage source → Advanced → select SINE.
- Enter the following values:

  DC offset (Voffset) = 0

  Amplitude (Vamp) = 1 ← normalised carrier

  Frequency = 20k ← fc = 20 kHz

  Tdelay = 0, Theta = 0, Phi = 0
- Right-click on the label V1 and rename it to Vc.
- Press 'G' → place ground below Vc. Connect the negative (−) terminal of Vc to Ground.
- Press 'W' to draw a wire from the positive (+) terminal of Vc.
- Press 'N' → assign Net Name: type 'carrier' and press OK. Paste it on the free wire.
- Hover over the wire — the bottom status bar must show 'carrier' (not N001 or N002).

Step A2: Place the Binary Message Source Vmsg (PULSE, 1 kbps)

- Select voltage source and place it to the right of Vc.
- Right-click on the voltage source → Advanced → select PULSE.
- Enter the following values:

  Vinitial = 0 ← bit-0 level

  Von = 1 ← bit-1 level

  Trise = 10n, Tfall = 10n ← fast transitions

  Ton = 0.5m ← bit period high = 0.5 ms

  Tperiod = 1m ← bit period = 1 ms (bit rate = 1 kbps)
- Right-click on the label and rename it to Vmsg.
- Press 'G' → connect the negative (−) terminal of Vmsg to Ground.
- Press 'W' → draw a wire from the positive (+) terminal of Vmsg.
- Press 'N' → assign Net Name: type 'msg' and press OK. Paste it on the free wire.

- Hover over the wire — the bottom status bar must show 'msg'.

Step A3: Place the Behavioral Voltage Source B1 (ASK Multiplier)

- Press 'P' → search bv → select Behavioral Voltage → place it to the right of Vc and Vmsg.
- Right-click B1 → in the Value field type EXACTLY:

  V = V(msg) * V(carrier)

B1 Expression — ASK Verification:

V(msg) = m(t) ∈ {0, 1} ← from PULSE source Vmsg (Step A2)

V(carrier) = sin(2π·fc·t) ← from SINE source Vc with Vamp=1 (Step A1)

B1 output = V(msg) × V(carrier)

= m(t) × sin(2π·fc·t)

= $s_{ASK}(t)$ ✔ matches standard ASK (OOK) equation

Step A4: Place Load Resistor R1 and Complete the ASK Circuit

- Press 'R' → place resistor. Right-click → set value to 50.
- Press 'W' → connect B1 positive (+) terminal to the top of R1.
- Press 'N' → label this wire: ASK_out.

  (Note: include your roll number in the label e.g. ASK_out_26)
- Press 'G' → ground B1 negative (−) terminal and the bottom of R1.

Step A5: Set Up and Run Transient Simulation (ASK)

- Click Simulate → Edit Simulation Cmd → Transient tab.
- Set Stop Time: .tran 5m ← 5 milliseconds = 5 complete message bits at 1 kbps.
- Place this directive on the schematic canvas. Click Run.

Step A6: View ASK Waveforms

- Click on the ASK_out wire → adds V(ASK_out) to the plot.
- Click on the msg wire → adds V(msg) to see the binary message.
- Click on the carrier wire → adds V(carrier) to see the unmodulated carrier.

- Observe: carrier present when msg = 1; carrier absent (zero output) when msg = 0.

Part B: FSK Circuit

***Step B1:* Place the Space Frequency Source Vf1 (SINE, 10 kHz)**

- Select voltage source and place it on the schematic.
- Right-click → Advanced → select SINE.
- Enter the following values:

  Amplitude (Vamp) = 1

  Frequency = 50k ← f1 = 10 kHz (space, bit-0)

  Voffset = 0, Tdelay = 0, Theta = 0, Phi = 0
- Rename the source to Vf1.
- Connect the negative (−) terminal to Ground.
- Assign Net Name 'f1' to the wire from the positive (+) terminal.

***Step B2:* Place the Mark Frequency Source Vf2 (SINE, 20 kHz)**

- Select voltage source and place it to the right of Vf1.
- Right-click → Advanced → select SINE.
- Enter the following values:

  Amplitude (Vamp) = 1

  Frequency = 100k ← f1 = 20 kHz (space, bit-0)

  Voffset = 0, Tdelay = 0, Theta = 0, Phi = 0
- Rename the source to Vf2.
- Connect the negative (−) terminal to Ground.
- Assign Net Name 'f2' to the wire from the positive (+) terminal.

***Step B3:* Place the Binary Message Source Vmsg (PULSE, 1 kbps)**

- Repeat Step A2 exactly. Use the same PULSE settings.

- Assign Net Name 'msg' to the wire from the positive (+) terminal.

Step B4: Place the Behavioral Voltage Source B1 (FSK Selector)

- Press 'P' → search bv → select Behavioral Voltage → place it.
- Right-click B1 → in the Value field type EXACTLY:

  V = IF(V(msg)>0.5, V(f2), V(f1))

B1 Expression — FSK Verification:

V(msg) > 0.5 → bit-1 detected → B1 outputs V(f2) = sin(2π·20k·t) (mark)

V(msg) ≤ 0.5 → bit-0 detected → B1 outputs V(f1) = sin(2π·10k·t) (space)

= $s_{FSK}(t)$ ✔ matches standard FSK equation

Step B5: Place Load Resistor R1 and Complete the FSK Circuit

- Press 'R' → place resistor with value 50.
- Connect B1 positive (+) terminal to the top of R1.
- Assign Net Name 'FSK_out' to this wire.

  (Note: include your roll number in the label e.g. FSK_out_26)
- Ground B1 negative (−) terminal and the bottom of R1.

Step B6: Set Up and Run Transient Simulation (FSK)

- Click Simulate → Edit Simulation Cmd → Transient tab.
- Set Stop Time: .tran 5m ← 5 ms = 5 complete bits at 1 kbps.
- Place the directive on the schematic canvas. Click Run.

Step B7: View FSK Waveforms

- Click on the FSK_out wire → adds V(FSK_out) to the plot.
- Click on the msg wire → adds V(msg) to see the binary message.
- Click on the f1 wire → adds V(f1) to see the 10 kHz space carrier.
- Click on the f2 wire → adds V(f2) to see the 20 kHz mark carrier.
- Observe: lower frequency (10 kHz) during bit-0; higher frequency (20 kHz) during bit-1.

Part C: BPSK Circuit

Step C1: Place the Carrier Source Vc (SINE, 10 kHz, Vamp = 1)

- Repeat Step A1 with Freq = 10 kHz.
- Assign Net Name 'carrier' to the wire from the positive (+) terminal.

Step C2: Place the Binary Message Source Vmsg (PULSE, 1 kbps)

- Repeat Step A2 exactly.
- Assign Net Name 'msg' to the wire from the positive (+) terminal.

Step C3: Place the Behavioral Voltage Source B1 (BPSK Phase Mapper)

- Press 'P' → search bv → select Behavioral Voltage → place it.
- Right-click B1 → in the Value field type EXACTLY:

  V = V(carrier) * (2*V(msg) - 1)

B1 Expression — BPSK Verification:

When V(msg) = 1 (bit-1): $2\times1 - 1 = +1$ → B1 = +1 × V(carrier) → 0° phase

When V(msg) = 0 (bit-0): $2\times0 - 1 = -1$ → B1 = −1 × V(carrier) → 180° phase reversal

B1 output = $(2\times m(t) - 1) \times \sin(2\pi \cdot fc \cdot t)$

= $s_{BPSK}(t)$ ✔ **matches standard BPSK equation**

Step C4: Place Load Resistor R1 and Complete the BPSK Circuit

- Press 'R' → place resistor with value 50.
- Connect B1 positive (+) terminal to the top of R1.
- Assign Net Name 'PSK_out' to this wire.
- Ground B1 negative (−) terminal and the bottom of R1.

Step C5: Set Up and Run Transient Simulation (BPSK)

- Click Simulate → Edit Simulation Cmd → Transient tab.
- Set Stop Time: .tran 5m ← 5 ms = 5 complete bits at 1 kbps.

- Place the directive on the schematic canvas. Click Run.

Step C6: View BPSK Waveforms

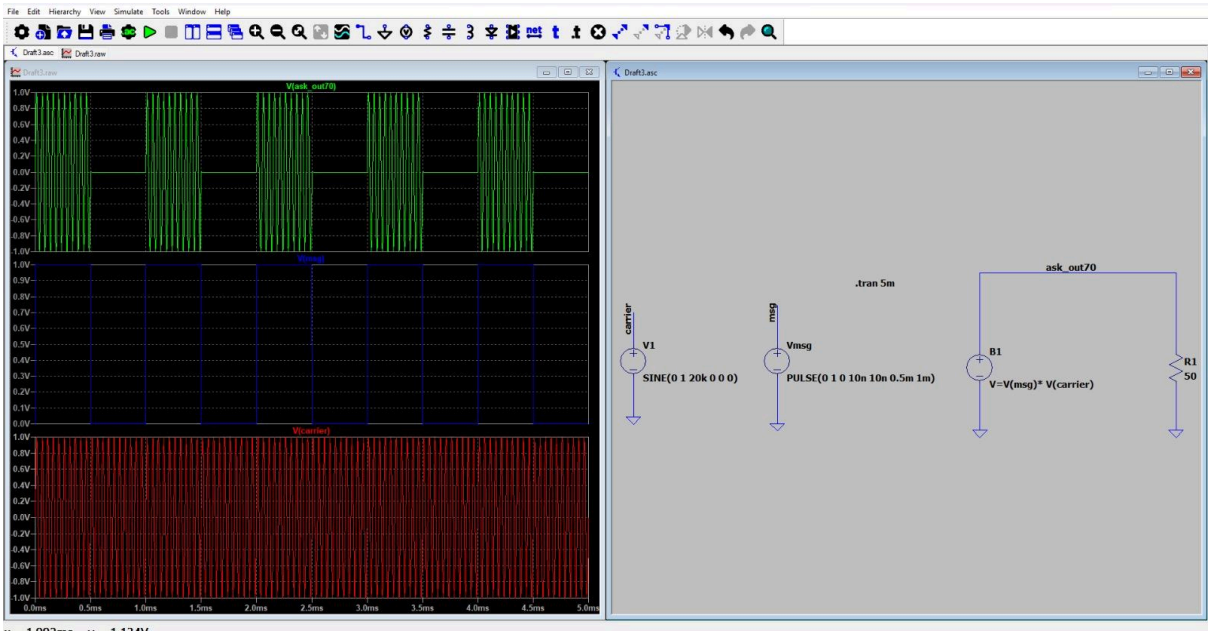
- Click on the PSK_out wire → adds V(PSK_out) to the plot.

(Note: include your roll number in the label e.g. PSK_out_26)

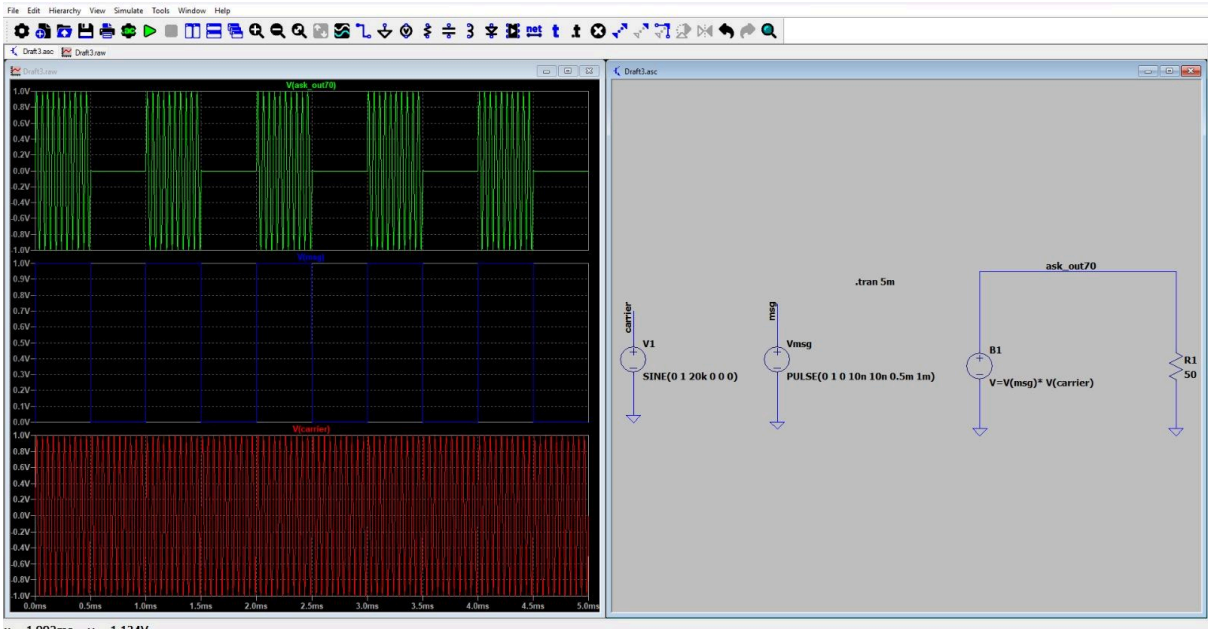
- Click on the msg wire → adds V(msg) to see the binary message.
- Click on the carrier wire → adds V(carrier) to see the unmodulated carrier.
- Observe: carrier phase reversal (180° shift) at every 0→1 or 1→0 transition in the message.

Observations

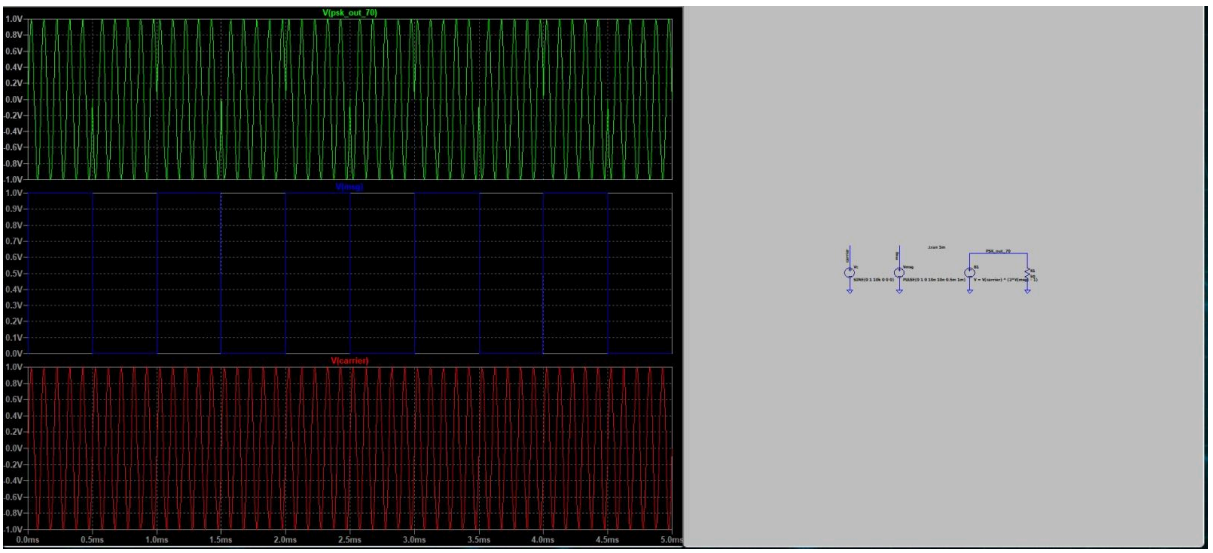
Part A (ASK):



Part B (FSK):



Part C (BPSK):



Modulation	Message Bit Pattern	Carrier Freq. / Freqs.	Bit Rate	Observable Feature	Observation (describe waveform)	Verified?
ASK	0V / 1V PULSE	fc = 20 kHz	1 kbps	Carrier on/off with bit	The output waveform **V(ask_out_58)** shows the **20 kHz** carrier sine wave appearing only when the message signal is at **1V**. When the message signal drops to **0V**, the carrier is completely suppressed, showing a flat line at **0V**. This confirms the implementation of **On-Off Keying (OOK)**.	Yes

Modulation	Message Bit Pattern	Carrier Freq. / Freqs.	Bit Rate	Observable Feature	Observation (describe waveform)	Verified?
FSK	0V / 1V PULSE	f1=10k, f2=20k	1 kbps	Frequency switches with bit	The output waveform **V(fsk_out_58)** shows a continuous carrier that switches between two distinct frequencies based on the message signal. When **V(msg)** is at **1V**, the output follows the **100 kHz** signal (f_2), and when the message is at **0V**, it switches to the **50 kHz** signal	Yes

Modulation	Message Bit Pattern	Carrier Freq. / Freqs.	Bit Rate	Observable Feature	Observation (describe waveform)	Verified?
					(f_1). The constant amplitude throughout confirms the successful implementation of **Frequency Shift Keying (FSK)**.	
BPSK	0V / 1V PULSE	fc = 10 kHz	1 kbps	180° phase reversal at transitions	The output waveform **V(psk_out_58)** maintains a constant amplitude and frequency (**20 kHz**), but its phase reverses by **180°** whenever the message signal	Yes

Modulation	Message Bit Pattern	Carrier Freq. / Freqs.	Bit Rate	Observable Feature	Observation (describe waveform)	Verified?
					transitions. When **V(msg)** is **1V**, the output is in phase with the carrier; when it is **0V**, the waveform is inverted. This distinct "phase flip" at the bit boundaries confirms the implementation of **Binary Phase Shift Keying (BPSK)**.	

Conclusion

Three digital modulation schemes were simulated in LTspice using a common PULSE voltage source (Vmsg, 0V/1V, 1 kbps) and Behavioral Voltage Sources (BV) as modulators.

- **ASK (OOK):** The BV expression V = V(msg) × V(carrier) directly implements On-Off Keying. The 20 kHz carrier was present when the message bit was 1 and absent when the bit was 0. The modulation was confirmed from the V(ASK_out) waveform.
- **FSK:** The BV expression V = IF(V(msg)>0.5, V(f2), V(f1)) switched between the 10 kHz space carrier (bit-0) and the 20 kHz mark carrier (bit-1). The frequency transition was clearly visible at each bit boundary in the V(FSK_out) waveform.
- **BPSK:** The BV expression V = V(carrier) × (2×V(msg) − 1) mapped bit-1 to +1 (0° phase) and bit-0 to −1 (180° phase reversal). The phase inversion at each 0↔1 and 1↔0 transition was clearly observable in the V(PSK_out) waveform.

All three circuits confirm the fundamental principle of digital modulation: a property of the carrier (amplitude, frequency, or phase) is varied in accordance with the binary message data.